Mathematics: analysis and approaches
Standard level
Paper 1
2.1–2.5 Functions

1 November 2026

W4ss Academy practice set 1 | **Topic** 2.1–2.5 Functions | **Ref.** W4SS–MAA–T2–P1–01

Student name

30 minutes

Instructions to students

- Write your name in the boxes above.
- Do not open this examination paper until instructed to do so.
- You are **not** permitted access to any calculator for this paper.
- This paper has a Section A only. Answer all questions. Answers must be written within the answer boxes provided.
- Unless otherwise stated in the question, all numerical answers should be given exactly or correct to three significant figures.
- A clean copy of the **mathematics: analysis and approaches formula booklet** is required for this paper.
- The maximum mark for this examination paper is **[25 marks]**.

Full marks are not necessarily awarded for a correct answer with no working. Answers must be supported by working and/or explanations. Where an answer is incorrect, some marks may be given for a correct method, provided this is shown by written working. You are therefore advised to show all working.

Section A

Answer **all** questions. Answers must be written within the answer boxes provided. Working may be continued below the lines, if necessary.

1. [Maximum mark: 5]

The following table shows values of $f(x)$ and $g(x)$ for different values of x.

Both f and g are one-to-one functions.

x	-2	0	3	4
$f(x)$	8	4	0	-3
$g(x)$	-5	-2	4	0

(a) Find $g(0)$. [1]

(b) Find $(f \circ g)(0)$. [2]

(c) Find the value of x such that $f(x) = 0$. [2]

2. [Maximum mark: 5]

Consider the points A(−2, 20), B(4, 6) and C(−14, 12). The line L passes through the point A and is perpendicular to [BC].

(a) Find the equation of L. [3]

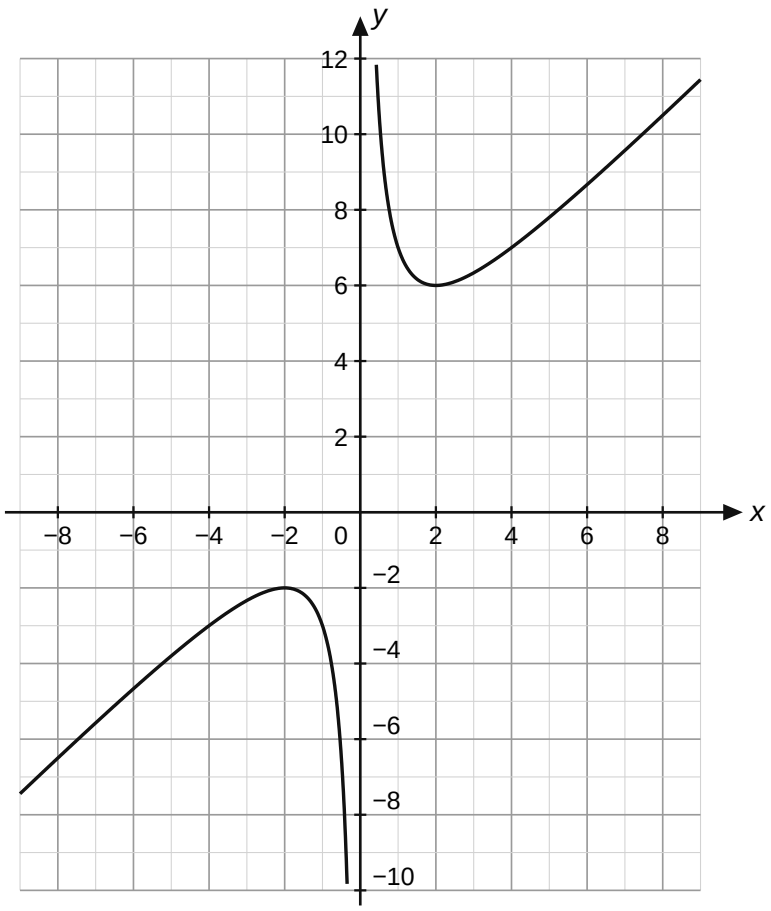
(b) The line L passes through the point $(k, 2)$.

Find the value of k. [2]

3. [Maximum mark: 4]

The function f is of the form $f(x) = ax + b + \frac{c}{x}$, where a , b and c are positive integers.

Part of the graph of $y = f(x)$ is shown on the axes below. The graph of the function has its local maximum at $(-2, -2)$ and its local minimum at $(2, 6)$.



(a) (i) Draw the line $y = -6$ on the axes. [1]

(ii) Write down the number of solutions to $f(x) = -6$. [1]

(b) Find the range of values of k for which $f(x) = k$ has no solution. [2]

(This question continues on the following page)

(Question 3 continued)

4. [Maximum mark: 5]

The functions f and g are defined such that $f(x) = \frac{x+3}{4}$ and $g(x) = 8x + 5$.

(a) Show that $(g \circ f)(x) = 2x + 11$. [2]

(b) Given that $(g \circ f)^{-1}(a) = 4$, find the value of a. [3]

5. [Maximum mark: 6]

The function f is defined by $f(x) = 2x^3 + 5, -2 \leqslant x \leqslant 2$.

(a) Write down the range of f. [2]

(b) Find an expression for $f^{-1}(x)$. [2]

(c) Write down the domain and range of f^{-1}. [2]

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(Question 5 continued)

Sources of the questions

This practice paper was put together by W4ss Academy for internal class use. The questions are taken from the IB DP Mathematics: analysis and approaches Questionbank (standard level, topic 2 Functions) and cover subtopics 2.1 to 2.5. Some of them come from former courses (Mathematics SL and HL, Mathematical studies SL). They remain the property of the International Baccalaureate Organization and must not be published or sold.

Section A — question 1	22M.1.SL.TZ2.1 · 2.5 composite functions from a table
Section A — question 2	22M.1.SL.TZ1.1 · 2.1 gradient, perpendicular lines, equation of a line
Section A — question 3	17M.1.SL.TZ1.T_12 (former Mathematical studies SL course) · 2.2, 2.4 reading a graph, number of solutions of f(x) = k — parts (b) and (c) of the original, relabelled (a) and (b): part (a) is not in the source — figure redrawn by W4ss Academy from the data in the question
Section A — question 4	SPM.1.SL.TZ0.5 · 2.5 composite function and its inverse
Section A — question 5	17M.1.AHL.TZ2.H_2 (former Mathematics HL course) · 2.2 range · 2.5 inverse function, its domain and range

Please **do not** write on this page.

Answers written on this page
will not be marked.